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Software Engineer transitioning into AI Engineering, with hands-on experience building and deploying AI solutions across Agentic AI, RAG, Computer Vision and NLP. I bring established backend engineering experience together with practical skills in Python, FastAPI, LangChain, LangGraph, PyTorch, Hugging Face and Docker.

Projects	Visual Product Recommendation System Computer Vision & Similarity Search - Developed an end-to-end AI-powered visual product recommendation system enabling users to upload a fashion image and retrieve visually similar products from a 50,000-image vendor database. - Built a data-cleaning and preprocessing pipeline to validate images, identify invalid files, and remove duplicates. - Used a pretrained ResNet50 CNN as a feature extractor to generate 2,048-dimensional image embeddings. - Implemented FAISS vector similarity search using cosine similarity to efficiently retrieve and rank Top-K visually similar products. - Developed a FastAPI backend and interactive web interface supporting image uploads, configurable similarity thresholds, and ranked recommendations. - Containerised the application using Docker and deployed it on Hugging Face dataset & Spaces, with product images hosted through Hugging Face Datasets. Tech stack: Python, PyTorch, Torchvision, ResNet50, FAISS, NumPy, Pandas, Pillow, FastAPI, Pydantic, Jinja2, Uvicorn, HTML/CSS/JavaScript, Docker, Hugging Face Spaces/Datasets, Git/Git LFS, uv. Code URL: kpsajesh/visual-product-recommender Deployed URL: https://huggingface.co/spaces/kpsajesh/visual-product-recommender Radiology Report Summarisation Finetuning Google T5 model - Developed an end-to-end NLP solution to generate concise clinical impressions from free-text radiology findings. - Processed and cleaned a chest X-ray radiology report dataset for model training. - Fine-tuned a T5-small sequence-to-sequence Transformer using PyTorch and Hugging Face. - Evaluated model performance using ROUGE, achieving a ROUGE-1 score of 0.5 against a target of 0.3. - Built and validated a REST API using FastAPI for model inference. - Containerised the application using Docker. - Developed a lightweight HTML frontend for user interaction. - Deployed the complete inference application on Hugging Face Space. Tech Stack: Python, Pandas, Scikit-learn, PyTorch, Hugging Face Transformers/Datasets, T5-small, NLP, ROUGE, FastAPI, Pydantic, HTML/JavaScript, Docker, Git/GitHub, Hugging Face Spaces.
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	Code URL: https://github.com/kpsajesh/Summarisation Deployed URL: https://kpsajesh-medical-summerisation.hf.space/ GRAPH TESTER AI Agent Designed and developed a backend AI agent system using LangGraph with human-in-the-loop workflows, retry logic, and fallback mechanisms. Implemented NLP-driven decision-making using LLMs (OpenAI, Ollama). Improved system reliability through error handling and agent orchestration. https://github.com/kpsajesh/GraphTester - Generates testcases from requirements and posts to the Testrail via API - Tests the testcases with playwright and reads the playwright json bug report - Posts bugs to Jira via API with basic validations to avoid duplication - Adds details to the slack channel via api Technologies used: Python, ollama client, open AI API, Agentic AI, langchain & Lang graph LANGCHAIN_RAG Includes a RAG project for HR department using langchain RAG https://github.com/kpsajesh/Langchain_RAG Technologies used: Python, open AI API, langchain, RAG & Langgraph, FAISS, Chroma vector DB LENDING CLUB It is Deep learning project for a lending club for companies like Lending Club correctly predicting whether or not a loan will be a default is very important. In this project, using the imbalanced historical data from 2007 to 2015. https://github.com/kpsajesh/DeepLearning Technologies used: Python, Pandas, Deep neural network, Tensor flow Keras, Matplotlib, Seaborn, SMOTE, L2 regularisation & adam optimiser COHORTS OF SONGS This is an unsupervised machine learning project. The client aims to create cohorts of different songs to enhance song recommendations. These cohorts will be based on various relevant features, ensuring that each group contains similar types of songs. https://github.com/kpsajesh/MachineLearning Performed exploratory data analysis and cluster analysis to create cohorts of songs. The goal was to better understand the various factors that create a cohort of songs. Technologies used: Python, Pandas, Numpy, Machine Learning, KMeans Clustering & Agglomerative Clustering, Matplotlib, Seaborn, PCA, Silhouette score Myrus Learning Management System Project Description: This product has 2 applications; one is for eLearning & face to face learning and second one is for portfolio management. I am doing development of new features to the product mainly in backend, and covering support. Technologies used: php, .NET, C# ASP.NET, REST API, SQL Server 2019.
Technologies Known	• Langchain, LangGraph, Agentic AI, RAG, FastAPI • Deep Learning models with sequential and functional models in Tensor flow, keras, optimisers & regularization, pytorch, Convolutional Neural Network CNN, Recurrent Neural Network RNN, Transfer learning, hugging face transformer

	- Machine Learning Algorithms for regression, classification and clustering with Scikit-learn, imblearn, xgboost, catboost, vecstack - Natural Language Processing NLP, NLTK toolkit, Gensim, spacy, regular expressions, Audio processing MFCC and spectrogram, Textblob/ Vader/Flair, vector databases, text to speech and speech to text - Better understanding of machine learning concepts - supervised and unsupervised learning, hyper parameters etc. - Exploratory Data Analysis tools as Matplotlib, Seaborn and Plotly for ML development and to collaborate with data scientists - Python, Pandas, Numpy, Tensor flow and Keras - Azure, GCP, ML ops, LLM Ops - Docker, Kubernetes
Other known Technology Areas	- Developing web applications in C#.NET, VB.NET, MVC, RDBMS - SQL server and Oracle. - Python, Visual Studio with ASP.NET core, REST API, C#, JavaScript, Ajax, JSON, HTML - Git hub, scrum & agile methodologies, Jira, Toad. - Involved in requirement gathering, solution designing, stakeholder management, coding, code reviews and problem solving when working in Middle East, Oman - Prepare perform unit and integration testing, project documentations. - SDLC, scrum (agile) and DHL DePICT waterfall methodologies, Training, implementation
Conventional Tools	Back end & RDBMS : SQL Server, TSQL, Oracle Front end : Python, Visual Studio with ASP.NET core, REST API, C#, JavaScript, Ajax, JSON, HTML 5 Supporting Tools : Git hub, scrum & agile methodologies, Jira, Toad.
Education	- Bachelor of Technology (Electrical & Electronics), MG University, Kerala, India, equivalent to Bachelor (honours) degree / RQF Level 6 / SCQF Level 10 / CQFW Level 6 - Masters in Business Administration, Annamalai University India, equivalent to Master's degree RQF Level 7/SCQF Level 11/CQFW Level 7
Certifications	AI Engineer certification from Simplilearn with learning from Fundamentals to GenAI AI900 : Azure AI Fundamentals Scrum Fundamentals Certified (SFC™) & Scrum Product Owner Certification (SPOC™) from Scrum study, equivalent to PSPO and CSPO. DePICT project management from DHL, water fall.
Experiences	6 years of experience as software developer - Application Developer – Broadshield LLC, Stevenage, United Kingdom (from Dec 2022 onwards) - Senior Service Desk Analyst- Care UK - Software Developer to application owner – Bahwan Exel LLC, Muscat, Oman - Software Engineer - Barry Weh-miller Design Group, Chennai, India - Software Engineer - Proton Web PVT LTD

Achievements & Awards	• DHL CEO silver award 2020 in DSC MLEMEA region under sustainability for developing and implementing a digital assistant for energy saving. • Exceptional performance award from Bahwan Exel towards simplification of complex coastal operations using a VBA tool in a very small time frame, later it is added as modules to TMS software. • Selected as a member of BWIR star club, the highest recognition given to an employee based on revenue to company and individual performance in Barry-Wehmler International Resources.
Hobbies & Personal Projects	• Owner of a patent (intellectual property rights) for designing and developing a privacy protection system (partnered with spouse) for smart phone devices. • Passionate about electronics hobby circuits especially automated energy saving solutions. • Knowledge in personal finance management.
Personal Details and soft skills	• ILR Indefinite Leave to Remain, no sponsorship needed. • Completed 5 years in UK and eligible for SC clearance. • Very good understanding in personal finance and investments. • Excellent problem-solving skills and attention to detail, Team player and collaborative • Highly motivated, Knowledgeable with Data science, ML, DL, NLP & Gen-AI concepts, tools. • Effectively collaborate with different teams, team player, no compromise on quality, proactive approach, attention to detail, very good communication.